COMICCRAFT – AI COMIC STORY CREATOR USING GEMINI MODELS

3. Project Design Phase

3.1 Introduction

The Project Design Phase defines the overall structure and architecture of the ComicCraft application. It explains how the frontend, backend, AI models, image generation, comic panels, and PDF export are connected.

3.2 System Architecture

The main workflow of ComicCraft is:

User → Web Interface → FastAPI Backend → Gemini Models → Image Generation → Comic Layout → PDF Export

The user enters the required information through the web interface. The FastAPI backend receives the information and communicates with the AI services.

3.3 Frontend Design

The frontend is designed using:

- HTML
- CSS
- JavaScript
- Jinja2 templates

The interface provides input fields for story prompt, character name, setting, tone, and art style.

The generated comic is displayed through a comic preview page.

3.4 Backend Design

FastAPI is used as the backend framework. It handles:

- User requests
- Story generation
- AI model communication
- Image generation
- Comic panel processing

- PDF export
- API endpoints

3.5 AI Design

Google Gemini models are used for the main AI tasks.

The system can use one model to generate the basic comic outline and another model to generate detailed narration, dialogues, and image prompts.

3.6 Comic Panel Design

The generated story is divided into multiple comic panels.

Each panel can contain:

- Scene description
- Character information
- Narration
- Dialogue
- Image prompt
- Generated image

3.7 Project Structure

The project is organized into different folders such as:

- `app`
- `services`
- `templates`
- `static`
- `tests`
- `docs`

This structure makes the application easier to develop, test, and maintain.

3.8 Conclusion

The Project Design Phase provides a clear architecture for connecting the user interface, backend, Gemini models, image generation, comic layout, and PDF export into one complete application.